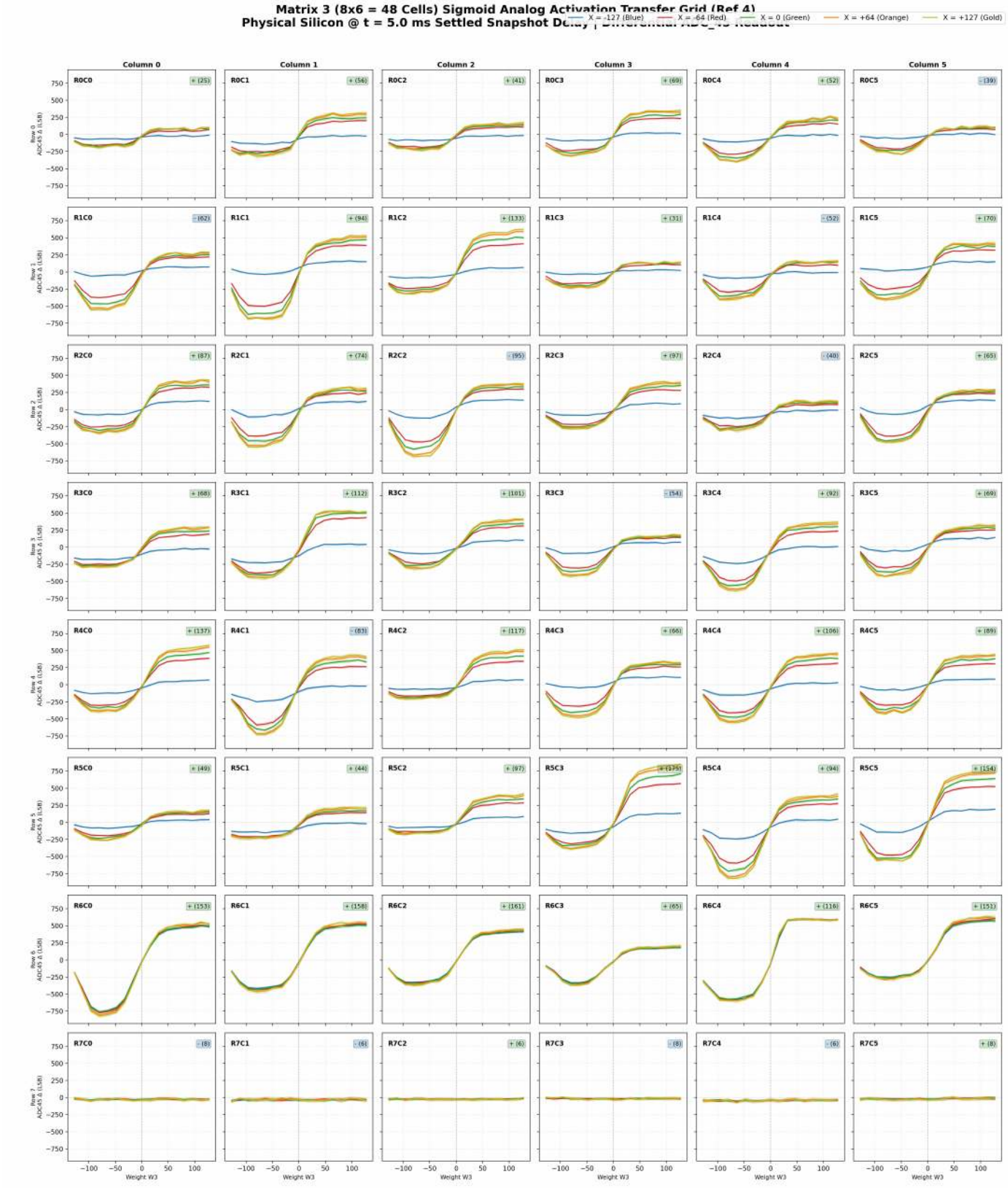
MATRIX 3 SIGMOID ANALOG ACTIVATION CHARACTERIZATION

Steg 1: Full 48-Crosspoint Transfer Landscape via ColTIA + Sigmoid (Ref 4) on Differential ADC_45
Pure Physical Silicon Telemetry @ Nominal t = 5.0 ms Settled Delay

EXECUTIVE SUMMARY & HARDWARE CONTEXT:
This empirical report characterizes all 48 crosspoints (8 rows × 6 cols) of Matrix 3 (Layer 1 adaptive compensation multiplier) driving the ColTIA summing stages and physical analog sigmoid activation circuitry (Ref 4). Measured via differential ADC_45 at nominal *t = 5.0 ms*. Each cell was excited across 17 weight levels [-127..+127] and 5 DAC_2 input amplitudes (X ∈ {-127, -64, 0, +64, +127}). The objective is to discover whether the positive (+) or negative (-) weight polarity offers superior dynamic swing and lower stochastic jerk, in order to set optimal initial baseline weights and regularization attractors.

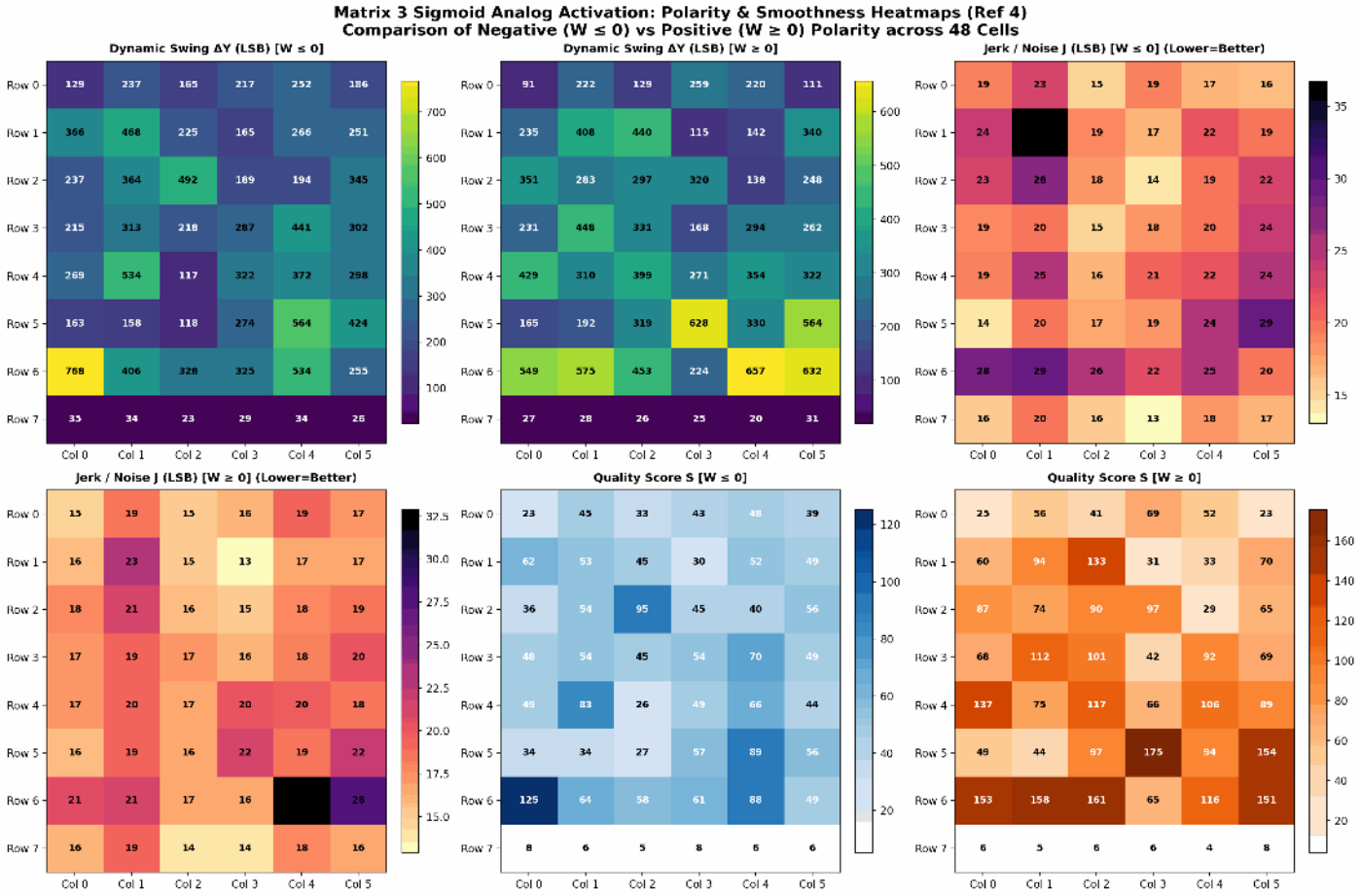
1. Full 48-Cell Transfer Grid (5 Input Curves per Cell)

Every cell transfer curve shows ADC_45 Δ (LSB) vs Weight W3. Curves: Blue (X=-127), Red (X=-64), Green (X=0), Orange (X=+64), Gold (X=+127). Shaded bands represent ±1σ repeatability across multiple synchronous recharges.

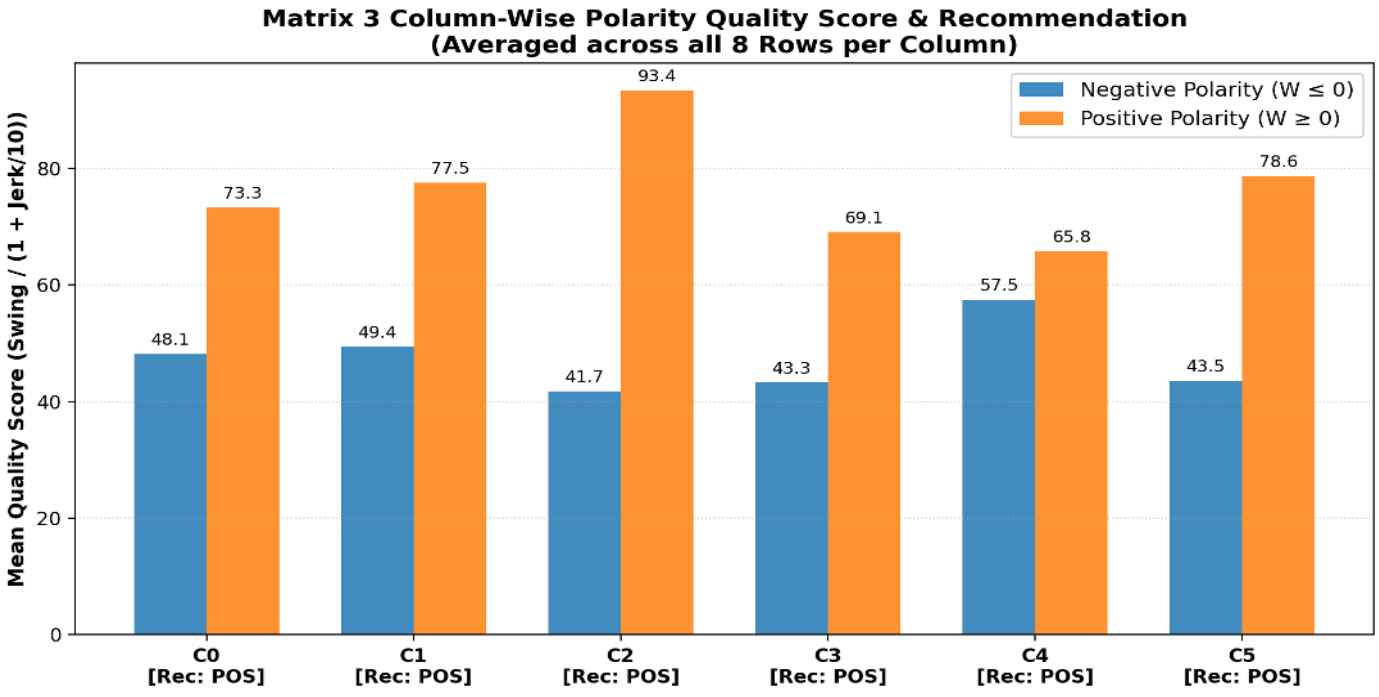

Matrix 3 (8x6 = 48 Cells) Sigmoid Analog Activation Transfer Grid (Ref 4)
X = -127 (Blue)
X = -64 (Red)
X = 0 (Green)
X = +64 (Orange)
X = +127 (Gold)
Column 0
Column 1
Column 2
Column 3
Column 4
Column 5
R0C0 + (25)
R0C1 + (56)
R0C2 + (41)
R0C3 + (69)
R0C4 + (52)
R0C5 - (39)
R1C0 - (62)
R1C1 + (94)
R1C2 + (133)
R1C3 + (31)
R1C4 - (52)
R1C5 + (70)
R2C0 + (87)
R2C1 + (74)
R2C2 - (95)
R2C3 + (97)
R2C4 - (40)
R2C5 + (65)
R3C0 + (68)
R3C1 + (112)
R3C2 + (101)
R3C3 - (54)
R3C4 + (92)
R3C5 + (69)
R4C0 + (137)
R4C1 - (83)
R4C2 + (117)
R4C3 + (66)
R4C4 + (106)
R4C5 + (89)
R5C0 + (49)
R5C1 + (44)
R5C2 + (97)
R5C3 + (175)
R5C4 + (94)
R5C5 + (154)
R6C0 + (153)
R6C1 + (158)
R6C2 + (161)
R6C3 + (65)
R6C4 + (116)
R6C5 + (151)
R7C0 - (8)
R7C1 - (6)
R7C2 + (6)
R7C3 - (8)
R7C4 - (6)
R7C5 + (8)
Weight W3

2. Polarity, Dynamic Swing, and Smoothness Heatmaps

Comparison of Negative Polarity (W ≤ 0) vs Positive Polarity (W ≥ 0) across all 48 cells. Score combines dynamic swing with inverse jerk/roughness: Score = Dynamic / (1 + Jerk/10) × Monotonicity.



3. Column-Wise Quality Comparison & Polarity Decision



4. Quantitative Column Summary & Polarity Recommendation

Column	Score (W ≤ 0)	Score (W ≥ 0)	Recommended Polarity	Verdict / Interpretation
C0	48.1	73.3	**POS (+)**	Optimal for positive sigmoid knee biasing
C1	49.4	77.5	**POS (+)**	Optimal for positive sigmoid knee biasing
C2	41.7	93.4	**POS (+)**	Optimal for positive sigmoid knee biasing
C3	43.3	69.1	**POS (+)**	Optimal for positive sigmoid knee biasing
C4	57.5	65.8	**POS (+)**	Optimal for positive sigmoid knee biasing
C5	43.5	78.6	**POS (+)**	Optimal for positive sigmoid knee biasing

5. Architectural Conclusions & Transition to Steg 2

Transition to Steg 2 (Column Collective Sigmoid S-Curve Sweep):
With Steg 1 establishing cell-level polarity characteristics, Steg 2 will now sweep all rows of each column collectively (W3[0..7][c] ramped simultaneously) across representative operating levels (X = ±127 full-scale and X = ±64 arbitrary medium signal). This reveals the macroscopic sigmoid S-curve into ADC_45 under full multi-row current injection, identifying the exact baseline weight bias point W_base[c] where sensitivity and smoothness are maximized.